THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Temperature and its Measurement** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Thermometer A reads 0 °C in melting ice and 100 °C in boiling water. Thermometer B reads 2 °C and 98 °C. For the same hot water at actual 50 °C, B would most likely read: 1

(a) 48 °C **(b)** 52 °C **(c)** 54 °C **(d)** 50 °C

2. Rod A conducts heat twice as fast as rod B. If wax pins fall from A in 20 s from the heated end, the same pin on B would fall in about: 1

(a) 5 s **(b)** 40 s **(c)** 10 s **(d)** 20 s

3. The best conductor-insulator pair for a cooking pot and its handle is: 1

(a) wood and plastic **(b)** aluminium and plastic
(c) plastic and aluminium **(d)** glass and iron

4. Hot water at 80 °C is poured into three cups: metal, ceramic and plastic. After 5 minutes the water is coolest in the: 1

(a) ceramic cup **(b)** plastic cup **(c)** all equal **(d)** metal cup

5. If a 20 °C object touches a 60 °C object, which gets warmer? 1

(a) the 60 °C object **(b)** the 20 °C object **(c)** both cool **(d)** neither

6. A student heats a metal rod and a glass rod at one end for 1 minute. She touches the other ends. She will find: 1

(a) no change **(b)** metal rod hotter **(c)** both same **(d)** glass rod hotter

7. Which conclusion follows if the temperature of an object is not changing though heat is flowing in and out? 1

(a) object is hot **(b)** no heat flow
(c) object is cold **(d)** heat in equals heat out

8. A clinical thermometer reads 36.8 °C and 37.0 °C in two successive tests. This suggests: 1

(a) error **(b)** fever
(c) hypothermia **(d)** normal temperature with small variation

9. **Assertion (A):** The sense of touch is not a reliable way to judge temperature. 1
Reason (R): Our skin can be misled by what we touched before, so the same water may feel hot or cold.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A clinical thermometer is used to measure the temperature of a human body. **Reason (R):** A laboratory thermometer is used to measure the temperature of boiling water. Choose the correct option: 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two beakers contain 100 mL water at 70 °C and 700 mL water at 70 °C. Compare their temperatures and say if the bigger one has more heat. 2

12. Why do some thermometers use alcohol coloured red instead of mercury, especially at low temperatures? 2

13. Which has more heat: a bucket of water at 40 °C or a cup at 90 °C? Explain. 2

14. A metal spoon and a wooden spoon at room temperature feel different to touch. Does this mean they have different temperatures? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student measures the temperature of boiling water as 98 °C in a hill station and 100 °C in a plain. Suggest a reason. 3

16. Describe why the lid of a tiffin box should be airtight to keep food warm. 3

17. What are the possible errors in using a thermometer to read the temperature of a liquid? 3

OR

Plan a fair test to find which material keeps water hot longest: wool, cotton, aluminium foil. 3

18. Explain how a daily temperature chart can help predict the next day's weather (basic idea). 3

OR

Describe an activity to show the reliability issue of touch using two hands and three bowls and how to correct it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ananya in Hyderabad felt hot and weak. Her father took a clinical thermometer, shook it so that the mercury level came below 35 degrees C, and placed it under her tongue for about a minute. The reading was 38.5 degrees C. 5

(i) Why was the thermometer shaken first?
(ii) What is the normal body temperature?
(iii) In which unit is the temperature read here?
(iv) Does Ananya have a fever? Explain why the thermometer should not be held in hot water.

20. Explain how heat is transferred in a metal spoon placed in boiling soup and why the plastic spoon stays cool. 5

OR

Explain the sources of error while using a thermometer and give practical steps to reduce them. 5

END OF QUESTION PAPER — Total: 40 Marks